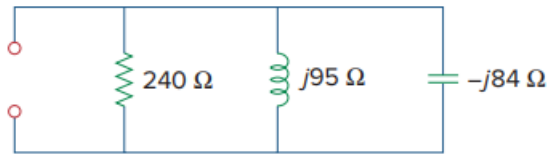


Problem

The circuit shown in Fig. 9.86 is used in a television receiver. What is the total impedance of this circuit?

Figure 9.86:



Step-by-step solution

Step 1 of 1

$$Y = \frac{1}{240} + \frac{1}{j95} + \frac{1}{-j84}$$

$$Y = 4.1667 \times 10^{-3} - j0.01053 + j0.0119$$

$$Z = \frac{1}{Y} = \frac{1000}{4.1667 + j1.37} = \frac{1000}{4.3861 \angle 18.2^\circ}$$

$$Z = 228 \angle -18.2^\circ\ \Omega$$